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9 February 1971
SUPERSEDING
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19 March 1963

MILITARY SPECIFICATION

SWITCHES, TOGGLE, ELECTRICALLY HELD, SEALED

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers general requirements for direct current electrically held, environmentally sealed toggle switches for use in alternating current (AC) and direct current (DC) applications. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Box, Set-Up
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, For Packaging
ZZ-R-765	Rubber, Silicone, Low and High-Temperature and Tear Resistant

Military

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting

FSC 5930

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SPECIFICATIONS

Military (Continued)

MIL-P-116	Preservation, Methods of
MIL-P-997	Plastic Material, Laminated, Thermosetting, Electric Insulation, Sheets, Glass-Cloth, Silicone Resin
MIL-W-5086	Wire, Electric, Hook-Up and Interconnecting, Polyvinyl Chloride-Insulated, Copper or Copper Alloy Conductor
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-P-15037	Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin
MIL-F-15160	Fuses, Instrument, Power, and Telephone
MIL-C-45662	Calibration System Requirements
MIL-I-81023	Inductor, 28 V, D.C. Laboratory Test, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-456	Electronic Parts, Date and Source Coding for

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STANDARDS

Military (Continued)

MS25017	Circuit Breaker, Trip-Free, Push-Pull, 5 to 50 Amp, Type I, Size A

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28	Screw-Thread Standards for Federal Services

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

SD-6	Provisions Governing Qualification

(Applications for copies should be addressed to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.1.2 Requirements for switches not covered by specification sheets. When specified (see 3.1 and 6.2.1), switches furnished under this specification, which are not covered by specification sheets, shall be products which will meet the requirements specified on the order or contract.

3.2 Qualification. Switches furnished under this specification, which are covered by specification sheets listed in supplement 1, shall be products which have been tested, have passed the qualification tests specified in 4.5, and have been listed on or approved for listing on the applicable Qualified Products List (see 6.3).

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3.3 Quality conformance. Switches furnished under this specification shall have demonstrated that it fulfills all of the requirements of this specification by satisfactorily performing all the quality conformance tests specified herein.

3.4 Material. Material shall be as specified herein. When a definite material is not specified, a suitable material shall be used which will enable the switches to conform to the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 Metals. All external metal parts, other than current-carrying parts, shall be of corrosion-resistant material, or shall be suitably protected to resist corrosion.

3.4.1.1 Ferrous materials. Ferrous material shall not be used for current-carrying parts except for header terminals.

3.4.1.2 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metals in contact, which tend toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy) is not acceptable. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. The use of dissimilar metals separated by a suitable insulating material is also permitted. Dissimilar metals are defined in MIL-STD-454, requirement 16.

3.4.2 Plastic material. Unless otherwise specified (see 3.1), molded plastic material shall conform to MIL-M-14, and laminated plastic material shall conform to MIL-P-997 or MIL-P-15037, however, laminated plastics shall not be used for the switch body. Other types of plastic materials possessing superior characteristics may be used provided the manufacturer submits acceptable evidence to the activity responsible for qualification during this qualification test program on the submitted product. In addition to these specifications, the plastic material shall not cause any hazard when operating under the extremes of environment and performance as specified herein. Such hazards may be defined as an tendency to support combustion, give off noxious gases in harmful quantities, give off any gases in quantities sufficient to cause explosion of sealed housing, give off any gases in a sealed housing that will cause contamination of the contact or other parts of the component, or form current-carrying tracks when subjected to arcing conditions encountered when any of the tests in this specification are performed on the component.

3.4.3 Elastomer seals. The material for all elastomer seals shall be silicone rubber in accordance with ZZ-R-765.

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3.5 Design and construction. Switches shall be so constructed as to insure proper operation when mounted in any position. The switches shall be of the design, construction, and physical dimensions specified (see 3.1).

3.5.1 Actuating lever. The actuating lever shall be metal with a lusterless finish and shall be insulated from all current-carrying parts.

3.5.2 Terminals. Terminals shall be as specified (see 3.1).

3.5.2.1 Solder terminals. Solder terminals shall be treated to facilitate soldering. Coatings such a hot solder, hot-tin dip, or gold plating are acceptable. When gold plating is used, its thickness shall be a minimum of 30 millionths of an inch. Gold wash or water lacquer shall not be used as a coating.

3.5.2.2 Screw terminals. Screw terminals shall be provided with assembled hardware as specified (see 3.1). Screw terminals shall be treated to facilitate soldering.

3.5.2.3 Wire-lead terminals. Wire-lead terminals shall be No. 20 wire leads per MIL-W-5086 Type II marked with circuit identification and wire gauge (1-20, 2-20, etc.) per MIL-W-5088.

3.5.2.4 Unused terminals. Terminals which are not functionally required as indicated in the applicable specification sheet (see 3.1) shall not be supplied.

3.5.3 Mounting means.

3.5.3.1 Mounting bushing. The mounting bushing shall be as specified (see 3.1). The bushing shall be provided with a seal between the toggle lever and the switching mechanism. The body of the seal shall not protrude from the top of the mounting bushing with exception of the tapered joint or meniscus where the seal joins the toggle lever.

3.5.3.2 Mounting hardware. The mounting hardware shall be as specified (see 3.1). For direct Government procurement, all mounting hardware shall be assembled in the proper order as shown on the applicable specification sheet (see 3.1).

3.5.4 Screw threads. Screw threads on external threaded parts shall be in accordance with Handbook H28. Threading of nonmetallic parts shall not be permitted. Terminal thread engagement shall be at least two full threads.

3.5.5 Soldering. Where soldering is employed, it shall be in accordance with MIL-STD-454 except that excess flux shall be removed.

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3.6 Operating characteristics. When switches are tested as specified in 4.9.2, the toggle lever angular travel, hold-in point, drop-out point, coil resistance, and override force shall be as specified (see 3.1).

3.7 Strength of terminals. When switches are tested as specified in 4.9.3.1, 4.9.3.2, or 4.9.3.3, there shall be no breakage, loosening, rotation of terminals, or other damage which would affect subsequent operation of the switch.

3.8 Strength of actuating lever pivot and lever stop. When switches are tested as specified in 4.9.4, there shall be no malfunction, damage, breakage, or short circuit; the switch shall operate mechanically and electrically.

3.9 Dielectric withstanding voltage. When switches are tested as specified in 4.9.5 there shall be no arcing, flashover, or leakage current in excess of 1.0 milliamperes.

3.10 Mechanical endurance. When switches are tested as specified in 4.9.6, there shall be no mechanical damage and the switch must properly control a pilot lamp circuit for a total of 10 operations at the conclusion of the test.

3.11 Electrical overload and endurance. When switches are tested as specified in 4.9.7, each pair of switch contacts under test shall open and close the circuit in proper sequence during each cycle of the switch actuating member. There shall be no external evidence of malfunction or damage during or after the test. This includes deterioration, electrical discharge, overheating, or evidence of combustion. At the conclusion of the test, the switch shall be electrically and mechanically operable while controlling the test load, at the test environment and at room conditions.

3.12 Seal. When switches are tested as specified in 4.9.8, the leakage rate shall not exceed 1×10^{-6} standard atmosphere cubic centimeter per second (atm cc/sec).

3.13 Vibration. When switches are tested as specified in 4.9.9, there shall be no separation of closed contacts exceeding 10 microseconds or closure of open contacts exceeding 10 microseconds. There shall be no evidence of damage.

3.14 Shock. When switches are tested as specified in 4.9.10, there shall be no separation of closed contacts exceeding 10 microseconds or closure of open contacts exceeding 10 microseconds. There shall be no evidence of damage.

3.15 Acceleration. When switches are tested as specified in 4.9.11, there shall be no separation of closed contacts or closure of open contacts. There shall be no evidence of damage.

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3.16 Salt spray. When switches are tested as specified in 4.9.12, there shall be no warping, cracking, or corrosion which will adversely affect subsequent operation. The specified cycling shall be completed without failure. After the test, the mounting hardware shall be readily removable.

3.17 Moisture resistance. When switches are tested as specified in 4.9.13, there shall be no damage which would adversely affect subsequent operation.

3.18 Sand and dust. When switches are tested as specified in 4.9.14, the switch shall properly control a pilot lamp circuit for a total of 10 operations at the conclusion of the test.

3.19 Explosion. When switches are tested as specified in 4.9.15, there shall be no explosion. If the switch causes an explosion at any of the test altitudes, it shall be considered to have failed to pass the test and no further testing may be attempted.

3.20 Marking. Switches shall be marked in accordance with MIL-STD-130 with the manufacturer's name or trademark, the manufacturer's part number, and the applicable specification sheet part number. The date of manufacture shall be marked in accordance with MIL-STD-456. The terminals shall be permanently identified. The switch case shall be permanently marked with the switch circuit diagram as shown on the applicable specification sheet (see 3.1).

3.21 Workmanship. Switches shall be manufactured and processed in a careful and workmanlike manner in accordance with good design and sound practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Nothing specified herein shall prevent the supplier or manufacturer from taking additional samples and making such additional inspection as he may deem necessary or desirable to assure conformance of switches to this specification.

4.2 Classification of inspection. The examination and testing of switches shall be classified as follows

(a) Component-materials inspection (see 4.3).

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(b) Qualification inspection (see 4.5).

(c) Inspection requirements for switches not covered by specification sheets (see 4.6).

(d) Quality-conformance inspection (see 4.7).

(1) Inspection of product for delivery (see 4.7.1).

(2) Inspection of preparation for delivery (see 4.8).

4.3 Component-materials inspection. Component-materials inspection shall consist of verification that the component materials listed in Table I, used in fabricating the switches, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE I

COMPONENT-MATERIALS INSPECTION

Component Material	Requirement Paragraph	Applicable Specification
Plastic	3.3.2	MIL-M-14, MIL-P-997, MIL-P-15037
Rubber	3.3.3	ZZ-R-765

4.4 Inspection conditions. Unless otherwise specified herein, all tests required by this specification shall be made at room ambient conditions in accordance with the "general requirement" of MIL-STD-202.

4.4.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required acceptance tests. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government. Calibration of standards which control the accuracy of the inspection equipment shall comply with the requirements of MIL-C-45662.

4.5 Qualification inspection. Qualification inspection shall be performed at a Government laboratory or private test facility approved by the Government. Qualification inspection shall consist of the examinations and tests specified in Table II and in the order shown.

4.5.1 Sample. For qualification inspection, the sample size for each test shall be as specified in Table II.

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TABLE II

QUALIFICATION INSPECTION

Examination or Test	Requirement Paragraph	Method Paragraph	Sample Number																					
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Examination of Product	3.1, 3 20, 3 21	4 9.1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Operating Characteristics	3 6	4.9 2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Dielectric Strength	3.9	4 9.5	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4	3 4
Sealing	9.12	4.9.8																						
Strength of Terminals	3.7	4 9 3	5	5																				
Strength of Actuating Lever	3.8	4.9 4	6	6																				
Moisture Resistance	3.17	4.9 13	7	7																				
Vibration	3.13	4.9 9			5	5																		
Shock	3.14	4.9.10			6	6																		
Acceleration	3.15	4.9 11			7	7																		
Salt Spray	3 16	4 9.12			8	8																		
Sand & Dust	3.18	4 9 14					5	5																
Explosion	3.19	4 9 15							5	5														
Overload	3.11	4 9 7.1									5	5	5	5	5	5	5	5	5	5	5	5	5	5
Electrical Endurance	3.11	4 9 7																						
Resistive DC	3 11	4 9 7 2									6	6												
Inductive DC	3.11	4 9 7.3											6	6										
Motor Load DC	3 11	4 9 7 4													6	6								
Resistive AC	3 11	4 9 7 2															6	6						
Inductive AC	3 11	4 9 7.3																	6	6				
Motor Load AC	3 11	4.9.7.4																			6	6		
Mechanical Endurance	3.10	4.9 6																					5	5
Sealing	9 12	4.9.8									7		7		7		7		7		7		6	6

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4.5.1.1 Single submission. A sample consisting of the required number of specimens of each switch shall be submitted. Qualification shall be restricted to the type submitted.

4.5.1.2 Group submission. The extent of qualification shall be in accordance with the applicable specification sheet (see 3.1)

4.5.1.3 Failures. Failure of any switch to comply with the applicable requirements shall be cause for refusal to grant qualification.

4.5.1.4 Test data. Each submission shall be accompanied by test data covering the tests listed in Table II which have been performed on representative specimens. The representative specimens shall be the types of switches being submitted for test. All test data shall be submitted in duplicate.

4.5.1.5 Certification of material. When submitting samples for qualification, the manufacturer shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

4.5.1.6 Description of items The manufacturer shall submit his part number for each switch for which qualification is sought.

4.6 Inspection requirements for switches not covered by specification sheets. Inspection requirements for switches not covered by specification sheets will be performed by the supplier, as specified in 4.7.1 after award of contract and prior to production, at a location designated by the Government. When specified, the samples and test routine shall be as specified in 4.5.1.1 and 4.5.1.2. Group B inspection is not applicable for switches not covered by specification sheets. Approval of switches not covered by specification sheets may be extended to a contract, other than the one under which it is granted.

4.7 Quality-conformance inspection.

4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Group A inspection.

4.7.1.1 Group A inspection. Group A inspection shall consist of the performance of tests specified in Table III in the order shown. Examination of product shall be limited to the inspection of marking and workmanship (see 4.9.1).

4.7.1.2 Sampling plan. Statistical sampling shall be in accordance with MIL-STD-105. The Acceptable Quality Level (AQL) shall be as defined in MIL-STD-105.

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TABLE III

GROUP A INSPECTION

Examination or Test	Method. Paragraph	AQL (percent defective)	
		Major	Minor
Seal	4.9.8	1/	
Examination of product	4.9.1	1.0	4.0
Operating characteristics	4.9.2	1.0	2/
Dielectric withstanding voltage	4.9.5	1.0	2/

1/ 100% inspection, no defects allowed.
2/ All defects in these categories are considered major defects.

4.7.1.3 Inspection lot. An inspection lot shall be as specified in MIL-STD-105 with the following modifications

(a) All switches that appear on the same specification sheet and that are offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.7.2 Group B inspection. Group B inspection shall consist of the tests specified in Table IV in the order shown.

TABLE IV

GROUP B INSPECTION

Examination of Test	Requirement Paragraph	Method Paragraph	1	2	3	4	5	6	7	8	9	10
Vibration	3.13	4.9.9	1	1								
Shock	3.14	4.9.10	2	2								
Mechanical endurance	3.10	4.9.6			1	1	1	1				
Overload	3.11	4.9.7.1							1	1	1	1
Electrical endurance	3.11	4.9.7										
Inductive DC	3.11	4.9.7.3							2	2	2	2
Sealing	3.12	4.9.8			3	3	3	3	3	3	3	3

4.7.2.1 Sampling plan. Group B inspection shall be performed on sample units selected from lots which have passed Group A inspection, unless the Government considers it more practical to select a sample from current production. The sample units shall be selected from those items covered by a single specification

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sheet. All samples shall be selected without regard to their quality. A manufacturer's normal quality control tests, production tests, environmental tests, and so forth, may be used to fulfill all or part of Group B inspection. All Group B inspection shall be completed as specified during each 36 month period following the initial qualification of the product.

4.7.2.2 Disposition of sample units. Sample units subjected to Group B inspection shall not be delivered on the contract or order.

4.7.2.3 Noncompliance. If a sample fails to pass Group B inspection the manufacturer shall take corrective action on the process and on all units of product which can be corrected and which were manufactured under essentially the same conditions, and with the same materials, processes, etc., and are considered subject to the same test failure. Quality conformance inspection shall be discontinued until corrective action has been taken. After the corrective action, sample units shall be subjected to the necessary Group B inspection (all inspection, or the failed inspections, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the Group B inspection has shown that the corrective action was successful.

4.7.2.4 Retention of qualification. To retain qualification, the manufacturer shall forward a summary of Group B tests performed during each 36 month period to the qualifying activity. Failure to submit the summary will result in loss of qualification approval for that product. When production for a particular type of switch has been suspended for 36 months or more, Group B tests shall be performed on the first lot produced when production of the item is resumed.

4.8 Inspection of preparation for delivery. Packages shall be inspected for conformance with Section 5 of this specification.

4.9 Methods of examination and test. Switch holding coils will not be energized unless otherwise specified.

4.9.1 Examination of product. Switches shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements. (See 3.1, 3.4, 3.5, 3.20, and 3.21.)

4.9.2 Operating characteristics (see 3.6). Switches shall be examined to determine that the following characteristics are within specified limits.

4.9.2.1 Toggle lever angular travel. Switches shall be mounted in a suitable fixture and toggle lever angular travel shall be measured.

4.9.2.2 Drop-out point. Switch coils shall be energized with maximum rated voltage, the toggle lever shall be transferred to the "electrically held"

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position. Coil voltage shall then be reduced at an approximate rate of one-half volt per second until the drop-out point is reached.

4.9.2.3 Hold-in point. Switch coil voltage shall be increased with the lever manually held in the momentary position until the hold-in point is reached.

4.9.2.4 Override force. The force required to override the holding coil shall be determined with the coil energized at maximum operating voltage. The force shall be applied at the tip of the actuating lever, perpendicular to the lever axis, and parallel to the line of travel.

4.9.2.5 Coil resistance. The resistance of the coil shall be measured by any suitable means

4.9.3 Strength of terminals (see 3.7).

4.9.3.1 Solder terminals. Solder terminals shall be subjected to a 5-pound pull for 1 minute in the following directions

(a) With the long dimension of the switch case vertical and mounting means horizontal, the specified pull shall be applied directly downward

(b) With the long dimension of the switch case horizontal and mounting means vertical, the specified pull shall be applied directly downward.

(c) With the long dimension of the switch case tipped 45° from horizontal, the specified pull shall be directly downward.

Any one terminal, however, shall be tested in one direction only.

4.9.3.2 Screw terminals. Screw terminals shall be subjected for 1 minute to:

(a) A 25-pound pull directly away from the switch body in each of the following directions.

(1) Parallel to the long axis of the terminal screw.

(2) Perpendicular to the long axis of the terminal screw.

Any one terminal, however, shall be tested in one direction only.

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(b) A torque of 6 pound-inches applied clockwise to the terminal screw in a plane parallel to the face of the terminal.

4.9.3.3 Wire-lead terminals. Wire-lead terminals shall be subjected to a 15-pound pull for 1 minute in any direction (including the direction most likely to cause failure).

4.9.4 Strength of actuating lever pivot and lever stop (see 3.8). The switch shall operate a pilot lamp circuit for 10 cycles of operation after the following tests:

(a) A 25-pound load shall be applied to the tip of the actuating lever for 1 minute under each of the following conditions

(1) Perpendicular to the lever axis and parallel to the line of lever travel at each end position of the lever. For lever-lock types, the test shall be conducted at each fixed position of the lever.

(2) Perpendicular to the lever axis and perpendicular to the line of travel at each lever position.

(3) Coaxial with the lever axis away from the lever pivot, throughout the entire range of lever travel.

(b) A 15-pound load shall be applied to the tip of the actuating lever, coaxial with the lever axis and toward the lever pivot, throughout the entire range of lever travel. For lever-lock types, this test is only applicable to those changes in lever position which may be accomplished without lifting the lever from its detent position.

4.9.5 Dielectric withstanding voltage (see 3.9). Switches shall be tested in accordance with Method 301 of MIL-STD-202. The following details shall apply:

(a) Test Voltage - 1000 volts rms, 60 cycles.

(b) Duration of Applications - 1 minute for qualification and Group B tests; 5 seconds for other tests.

(c) Points of Application -

(1) Between all terminals and the toggle lever.

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(2) Between all adjacent contacts of adjacent poles.

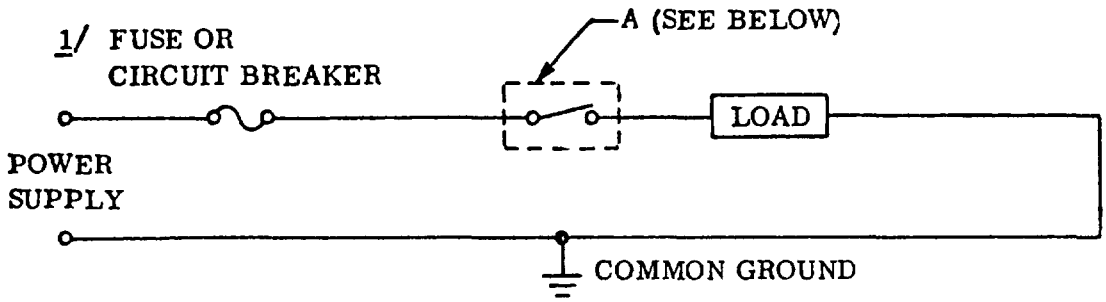
(3) Between all unconnected terminals of the same pole.

4.9.6 Mechanical endurance (see 3.10). The switches shall be subjected to 40,000 cycles of operation, 20,000 at -65 ±2° C, and 20,000 cycles at 71 ±2° C, at a rate of 50 to 60 cycles per minute. A cycle shall consist of movement of the toggle lever from one extreme position to the other extreme position and return to the original position. During testing of lever-lock switches, the pull-to-unlock mechanism may be by-passed to facilitate mechanical operation. In addition, pull-to-unlock switches shall be subjected to 5,000 cycles of pulling the toggle lever to the fully extended position and allowing it to return to the fully retracted position by internal spring action. Following the test the switch shall be examined and there shall be no mechanical breakage or malfunction as evidenced by its ability to control a pilot lamp circuit for 10 cycles of operation.

4.9.7 Electrical overload and endurance (see 3.11). Test switches shall be mounted by their normal mounting means against a metal plate. No insulator shall be interposed between the switch case and the metal plate. This metal plate, one side of power supply, one side of the test load, and the metal actuator of the actuating machine shall all be connected to a common ground. See Figure 1. In the case of a direct current circuit, the negative side of the power supply shall be grounded. No insulator shall be interposed between the grounded metal actuator and the switch actuating member. The test switch shall be connected between the power supply and the test load using a 1 to 3 foot length of wire of the size specified in Table V for connection to the switch. One throw of each pole of the test switch shall be connected to an independent test circuit meeting the specified conditions. Multipole switches are to be tested with all poles of the throw under test loaded. For double-throw switches, one half of the switches shall be tested with the load circuit connected to one throw and the remaining half of the switches shall be tested with the load circuit connected to the other throw. The switch shall be actuated at a frequency of 10 to 12 cycles per minute by a power driven metal actuator. During the overload test the closed time of the test switch shall be 0.5 second minimum duration during each cycle of operation. Unless otherwise specified the endurance test duty cycle shall be approximately 50-percent on, 50-percent off. Overload, resistive endurance, inductive endurance, and motor load endurance tests shall be performed at each voltage and electrical frequency specified on the detail specification. On each switch the overload and endurance test shall be performed on the same pair(s) of contacts, using the same voltage and electrical frequency for both tests. The endurance tests shall be performed in each case for 20,000 make and break cycles. For the first 10,000 cycles, the toggle lever shall be mechanically actuated, electrically held, and electrically released by a suitable auxiliary device. For the remaining 10,000 cycles, the toggle lever shall be mechanically actuated, electrically held, and mechanically overridden to effect release. (Holding coil to be energized at maximum rated voltage.) During testing of lever-lock switches,

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the pull-to-unlock mechanism may be by-passed to facilitate mechanical actuation. The test voltage magnitude and electrical frequency shall be maintained within the tolerance specified when the test voltage is measured at the switch terminals. Monitoring circuits shall be provided to detect failure of any pair of contacts to open or close their circuit in proper sequence. Monitoring circuits shall not shunt switch contacts or inductive components in the test circuit.



"A" = Metal mounting plate and metal actuator of actuating machine.

1/ See Table V for proper fuse or circuit breaker.

Figure 1. Test set-up for electrical endurance test

TABLE V

ENDURANCE TEST WIRE SIZE AND CIRCUIT BREAKER OR FUSE DESIGNATIONS

Endurance Test Current Amperes 1/	MIL-W-5088 Wire Sizes	Circuit Breaker or Fuse
3.0 or less	AN-20	MIL-F-15160/02, characteristic B, rating as applicable
5.0	AN-20	MS25017-5
7.5	AN-18	MS25017-7
10.0	AN-18	MS25017-10
15.0	AN-18	MS25017-15
18.0	AN-16	MS25017-20
20.0	AN-16	MS25017-20
25.0	AN-14	MS25017-25
30.0	AN-14	MS25017-30
40.0	AN-12	MS25017-35
60.0	AN-10	MIL-F-15160/10
80.0	AN-8	MIL-F-15160/6
175.0	AN-2	MIL-F-15160/9

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1/ Where the wire size, circuit breaker, or fuse size do not coincide with the required test current the next larger wire size, circuit breaker, or fuse shall be used.

4.9.7.1 Overload circuit. The switch shall make and break 150 percent of rated resistive load at the applicable voltage and electrical frequency for 50 cycles of operation at room conditions.

4.9.7.2 Resistive load (DC and AC). One half of the switches for test shall make and break the rated resistive load for 20,000 operating cycles. The other half of the switches for test shall make and break the rated resistive load for 20,000 operating cycles while at a pressure equal to 65,000 feet altitude.

4.9.7.3 Inductive load (DC and AC). One half of the switches shall make and break the rated inductive load for 20,000 operating cycles. The other half shall make and break the rated inductive load for 20,000 operating cycles while at a pressure equal to 65,000 feet. The DC inductive circuit shall be in accordance with MIL-I-81023. The AC inductive circuit shall have a power factor of 70 ±5 percent. The duty cycle for this test shall be 25 to 30 percent on and 70 to 75 percent off.

4.9.7.4 Motor load (DC and AC). Switches shall be subjected to 20,000 operations of making six times the rated motor load current and breaking the rated motor load current. The duration of the inrush current shall not be less than .05 seconds. The test load shall consist of only resistive components.

4.9.8 Seal (see 3.12). Switches shall be tested in accordance with Method 112 of MIL-STD-202. The following details shall apply

(a) Test condition letter - c.

(b) Procedure - I, Leakage rate not to exceed 1×10^{-6} atm cc/sec.

(c) Reduced pressure of chamber 1×10^{-4} torr approx.

(d) Detection exposure time - 1 second minimum.

(e) Gross leak check - not applicable.

4.9.9 Vibration tests (see 3.13).

4.9.9.1 Vibration. Switches and switch assemblies shall be tested in accordance with Method 204 of MIL-STD-202, as follows

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(a) Test condition letter - A. Total test duration shall be divided approximately equally between all toggle lever positions. When switches are tested in the "electrically held" position, the coil shall be energized with maximum rated voltage.

(b) Tests and measurements prior to vibration - Not applicable.

(c) Mounting - Switches shall be rigidly mounted by their normal mounting means on a rigid metal panel. The mounting fixture shall be free from resonances over the test frequency range.

(d) Tests and measurements during vibration - All open and closed circuits must be continuously monitored by a circuit capable of detecting contact opening or closing in excess of 10 microseconds duration.

(e) Tests and measurements after vibration - Shall be measured for operating characteristics (see 4.9.2).

(f) Examination after test - Switches and switch assemblies shall be examined for change in actuated position and evidence of broken, deformed, displaced, or loose parts.

4.9.10 Shock (see 3.14). Switches shall be tested in accordance with Method 213 of MIL-STD-202. The following details and exceptions shall apply

(a) Mounting - The switch shall be rigidly mounted by its normal mounting means.

(b) Test condition letter - A. One shock shall be applied in each toggle lever position in each direction of the three major axis of the switch. When switches are tested in the "electrically-held" positions, the coil shall be energized with maximum rated voltage.

The switches shall be continuously monitored for opening or closing of contacts in excess of 10 microseconds duration. After the test, operating characteristics shall be measured (see 4.9.2).

4.9.11 Acceleration (see 3.15). Switches shall be subjected to an acceleration force of 20 G attained within 1 minute. The force shall be maintained for 1 minute in each direction along each of the switches three major perpendicular axes. The switches shall be monitored for opening of closed contacts and closing of open contacts with a pilot lamp circuit. The test shall be repeated for each toggle lever position. When switches are tested in the "electrically-held" positions, the coil shall be energized with maximum rated voltage.

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4.9.12 Salt spray (see 3.16) Switches shall be tested in accordance with Method 101 of MIL-STD-202. The following details and exceptions shall apply·

(a) Salt solution - 5 percent.

(b) Test condition letter - B.

(c) Following a drying period of 6 hours in a forced draft oven at a temperature of approximately 57° C, switches shall make and break a pilot lamp circuit for 10 cycles of operation.

4.9.13 Moisture resistance (see 3.17). Switches shall be tested in accordance with Method 106 of MIL-STD-202. The following details and exceptions shall apply.

(a) Mounting - By normal mounting means on noncorrosive metal panel positioned 15° from the vertical. The mounting panel shall have appropriate mounting holes so that switches are at least 1 inch apart.

(b) Polarization - During steps 1 to 7 inclusive, a DC potential of 100 volts shall be applied between current-carrying parts and panel. Negative polarity shall be applied to the panel.

(c) Steps 7(a) and 7(b) are not applicable.

(d) Final measurements - Following a 24-hour period at 25 ±5° C and 50 ±5 percent relative humidity, operating characteristics shall be measured as specified in 4.9.2.

(e) Water - Distilled or deionized.

4.9.14 Sand and dust (see 3.18). Switches shall be tested in accordance with Method 110, Test Condition B, of MIL-STD-202. During the test, the switches shall be mechanically operated for 2,500 cycles. The cycling rate shall be 10 to 12 cycles of operation per minute. During the testing of lever-lock switches, the pull-to-unlock mechanism may be by-passed to facilitate mechanical operation. Following the test, the switch shall operate electrically and mechanically as evidenced by its ability to control a pilot lamp circuit for 10 cycles of operation.

4.9.15 Explosion (see 3.19). Switches shall be tested in accordance with Method 109 of MIL-STD-202. The following detail shall apply:

(a) Electrical load - Switches shall be operated at their maximum rated dc inductive current (see 3.1). During testing of

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lever-lock switches, the pull-to-unlock mechanism may be by-passed to facilitate mechanical operation.

5. PREPARATION FOR DELIVERY

5.1 1) Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2 and 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Switches shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. The procedure used shall not be injurious to the item.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Switches shall be adequately cushioned and individually packaged in accordance with MIL-P-116, Method IA-8. Switches, so packaged, shall be placed in a supplementary container conforming to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate packaging. Switches, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size and shape, shall be of minimum cube and tare, and shall contain multiples of 5-unit packages, not to exceed 100 packages or ten pounds. No intermediate packaging is required when the total quantity shipped is less than 100 units.

5.1.2 Level C. Switches shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This package may conform to the suppliers commercial practice for retail distribution when such meets the requirements of this level.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2 and 6.2.1).

5.2.1 Level A. The packaged items shall be packed in fiberboard containers conforming to PPP-B-636, weather resistant, style optional, special requirement. In lieu of the closure and waterproof requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, ends, edges, and manufacturer's joints with waterproof tape, 3 inches wide conforming to PPP-T-60, Class 1. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636. Metal banding shall not be used.

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5.2.2 Level B. The packaged items shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style and use requirements optional.

5.2.3 Level C. The package items shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This pack shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such conforms to the requirements of this level.

5.3 Marking (see 6.2 and 6.2.1). In addition to any special marking required by the contract or order, each unit package, intermediate and exterior containers shall be marked in accordance with MIL-STD-129.

5.4 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical items to the greatest extent possible.

6. NOTES

6.1 Intended use. Switches furnished under this specification are direct current electrically held, sealed toggle switches which are intended for use in controlling alternating current (AC) and direct current (DC) electrical loads.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of the applicable specification sheet (3.1).

(c) Levels of preservation and packaging, packing, and applicable marking (see Section 5).

6.2.1 Ordering data for switches not covered by specification sheets.

(a) Same as 6.2(a).

(b) See 3.1.2.

(c) Same as 6.2(c).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List (QPL),

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whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

6.3.1 Application for qualification. The activity responsible for the Qualified Products List (QPL) for MIL-S-5594(AS) is the Naval Air Systems Command. The Naval Air Test Center, Patuxent River, Maryland, has been designated by the Naval Air Systems Command as agent for the establishment of the QPL. Requests for information pertaining to and applications for qualification should be addressed to:

Commander
Naval Air Test Center
Patuxent River, Maryland 20670

6.4 Definitions

6.4.1 Maximum operating voltage. The maximum operating voltage is the maximum voltage at which the switch coil is designed to operate.

6.4.2 Normal operating voltage. The normal operating voltage is the voltage range in which the switch coil is expected to operate.

6.4.3 Hold-in voltage. Hold-in voltage is the voltage applied to the coil which causes the contact action of the switch to be changed from momentary to maintained.

(a) Hold-in point - Hold-in point is the voltage at which this change occurs (momentary to maintained). It is reached as the applied voltage is increased.

(b) Hold-in range The voltage range from hold-in point to maximum operating voltage is the hold-in range.

6.4.4 Drop-out voltage. Drop-out voltage is the voltage applied to the coil which causes the contact action of the switch to be changed from maintained to momentary.

(a) Drop-out point. Drop-out point is the voltage at which this change occurs (maintained to momentary). It is reached as the applied voltage is decreased.

(b) Drop-out range. The voltage range from drop-out point to zero applied voltage is the drop-out range.

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6.4.5 Pilot lamp circuit. The pilot lamp circuit consists of a power supply and pilot lamp. The source voltage at the power supply shall not exceed 28 VDC. The steady-state current rating of the pilot lamp shall not exceed the current rating of the switch.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

MIL-S-5594B(AS) SWITCHES, TOGGLE, ELECTRICALLY HELD, SEALED, CEN SPEC FOR

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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